

DAY 14 – NMAP BASICS

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1. What does Nmap stand for?

Nmap stands for Network Mapper. It is a network scanning and discovery tool used to identify hosts, open ports, services, and other information about systems in an authorized environment.

2. What is Nmap used for?

Nmap is used for network discovery, port scanning, service/version detection, and basic security assessment. Administrators can use it to understand what services are exposed on systems they are authorized to test.

3. What is network discovery?

Network discovery is the process of finding which hosts or devices are reachable on a network and learning basic information about them. It is often the first stage of an authorized network assessment.

4. What is a port scan?

A port scan checks TCP or UDP ports on a target to determine their state, such as open, closed, or filtered. It helps identify which network services may be listening.

5. What does an open port generally indicate?

An open port generally indicates that an application or service is listening and accepting network connections on that port. The port itself is not automatically a security vulnerability.

6. Does an open port automatically mean vulnerability? Explain.

No. An open port is not automatically a vulnerability. It simply shows that a service is reachable. Security depends on the service, its configuration, authentication, software version, exposure, and whether known weaknesses exist.

7. What does closed mean conceptually?

A closed port means the target is reachable, but no application is currently listening on that port. The host responded in a way that indicates the port is accessible but not serving a connection there.

8. What does filtered mean conceptually?

Filtered means Nmap cannot determine whether the port is open because a firewall, packet filter, or other network control is blocking or interfering with the probes.

9. What does nmap 127.0.0.1 do?

It scans the local machine (localhost) at IP address 127.0.0.1 using Nmap's default scan behavior. It is useful for safely learning Nmap because the target is the user's own system.

10. What does -p do?

The -p option specifies which port or ports Nmap should scan. For example, -p 22,80,443 requests a scan of ports 22, 80, and 443 on the authorized target.

11. What does -sV do?

The -sV option enables service and version detection. Nmap probes open ports to identify the service and, when possible, its software/version information.

12. What does -oN do?

The -oN option saves Nmap output in normal human-readable text format to a file. Example: `nmap -oN day14-scan.txt 127.0.0.1`

13. Why should scan results be documented?

Documentation creates a clear record of what was tested, when it was tested, what was observed, and what actions may be needed. It also supports repeatability, reporting, and responsible security work.

14. What is cybersecurity scope?

Cybersecurity scope defines the systems, IP addresses, applications, ports, time period, and activities that are authorized for an assessment. A clear scope prevents accidental testing of systems that are not permitted.

15. Why must authorization exist before scanning?

Scanning can generate traffic and may affect systems, trigger security alerts, or violate rules if performed without permission. Authorization establishes that the target and the planned activity are approved.

16. What is the difference between scanning and exploitation?

Scanning is primarily about discovering hosts, ports, services, and configurations. Exploitation attempts to take advantage of a vulnerability to obtain an effect such as unauthorized access. Scanning does not require exploitation.

17. Why is localhost a good beginner target?

Localhost is a safe beginner target because 127.0.0.1 refers to the local system. It allows students to practice basic Nmap commands without intentionally scanning public or third-party systems.

18. How do ports relate to services?

A network service listens on one or more ports so that clients can communicate with it. Common examples include SSH on TCP 22 and HTTPS on TCP 443, although services can be configured to use different ports.

19. Why are unexpected listening services worth reviewing?

An unexpected listening service increases the system's exposed attack surface. It should be identified and checked to confirm whether it is required, properly configured, patched, and appropriately restricted.

20. Explain a responsible Nmap assessment workflow.

A responsible workflow is: define the target and authorization; stay within scope; perform safe discovery; scan required ports; identify services/versions when appropriate; document results; review unexpected exposure; recommend remediation; and avoid exploitation unless it is separately authorized.

1. Check Nmap version

```
nmap --version
```

Purpose: confirms that Nmap is installed and shows the installed version.

```
rishabhraj@Rishabhs-MacBook-Air ~ % nmap --version
Nmap version 7.991 ( https://nmap.org )
Platform: arm-apple-darwin25.6.0
Compiled with: nmap-liblua-5.4.8 openssl-3.6.4 libssh2-1.11.1 libz-1.2.12 libpcap-1.10.6 nmap-libpcap-1.10.6 nmap-libnmap-2.0.18 t-1.18.0 ipv6
Compiled without:
Available nsock engines: kqueue poll select
```

2. Basic localhost scan

```
nmap 127.0.0.1
```

Purpose: performs a basic scan of the local machine.

```
rishabhraj@Rishabhs-MacBook-Air ~ % nmap 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-22 19:40 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000027s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
5000/tcp  open  upnp
7000/tcp  open  afs3-fileserver
8021/tcp  open  ftp-proxy
Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
```

3. Scan selected ports

```
nmap -p 22,80,443 127.0.0.1
```

Purpose: checks the specified TCP ports only.

```

rishabhraj@Rishabhs-MacBook-Air ~ % nmap -p 22,80,443 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-22 19:42 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000063s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds

```

4. Service/version detection

nmap -sV 127.0.0.1

Purpose: attempts to identify services and versions on discovered open ports.

```

rishabhraj@Rishabhs-MacBook-Air ~ % nmap -sV 127.0.0.1

Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-22 19:43 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000022s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
5000/tcp  open  rtsp
7000/tcp  open  rtsp
8021/tcp  open  tcpwrapped

2 services unrecognized despite returning data. If you know the service/version, please submit the following f
s at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port5000-TCP:V=7.991%I=7%O=9/22%Time=6AB28D09%P=arm-apple-darwin25.6.0%
SF:r(GetRequest,8F,"HTTP/1\1\0320403\0320Forbidden\r\nContent-Length:\03200\
SF:r\nServer:\0320AirTunes/960\13\1\1\r\nX-Apple-ProcessingTime:\03200\r\nX-
SF:Apple-RequestReceivedTimestamp:\032017368360\r\n\r\n")%r(RTSPRequest,8F,
SF:"RTSP/1\1\0320403\0320Forbidden\r\nContent-Length:\03200\r\nServer:\0320Ai
SF:rTunes/960\13\1\1\r\nX-Apple-ProcessingTime:\03200\r\nX-Apple-RequestRec

```

5. Save normal output

nmap -oN day14-scan.txt 127.0.0.1

Purpose: saves the scan result to a text file for documentation.

```

rishabhraj@Rishabhs-MacBook-Air ~ % nmap -oN day14-scan.txt 127.0.0.1

Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-22 19:50 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000025s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
5000/tcp  open  upnp
7000/tcp  open  afs3-fileserver
8021/tcp  open  ftp-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds

```